Stiffness log sheet

Same distance for every piece; stiffer piece = more force and more signal in the skin. Fill one row per photo / step, top to bottom.

- **Stop a piece as soon as force passes ~30 N**, even before 2.0 mm. Write "stopped" in Notes and skip its remaining rows.
- Pieces at room temperature (don't hold them in your hand before pressing).
- Tare the MARK-10 with nothing touching before each piece. Force display in N, real-time mode (not peak).
- R = rigid reference piece (PLA or aluminium) with the same face as P1–P4.

Session

Field	Value
Date	
Skin ID	
Room temperature at start (°C)	
Room temperature at end (°C)	
Stripe period → px/mm	
Camera: exposure / WB / focus	

Pieces

Piece	Material / what it is	Face shape and size (mm)	Height (mm)	Expected (soft → hard)
P1				
P2				
P3				
P4				
R	rigid reference			rigid

Check presses (flat Ø10 mm, centre, 1 mm)

When	Force (N)	Max phase	OK?
Start 1			

When	Force (N)	Max phase	OK?
Start 2			
Start 3			
End			

Part A — On the skin (photo + force)

Rows 26–30: P1 pressed again at the end, to see how much the same piece varies.

#	Piece	Repeat	Depth (mm)	File name	Force (N)	Notes
1	P1	1	ref	P1_r1_ref.png	0.5	
2	P1	1	0.5	P1_r1_0.5mm.png	4.0/4.5	
3	P1	1	1.0	P1_r1_1.0mm.png	9.0/9.5	
4	P1	1	1.5	P1_r1_1.5mm.png	14/14.5	
5 Extra extra	P1 P1 P1	1 1 1	2.0 2.5 3.0	P1_r1_2.0mm.png P1_r1_2.5mm.png P1_r1_3.0mm.png	20/20.5 27/27.5 33.5/34	
6	P2	1	ref	P2_r1_ref.png	0.5	
7	P2	1	0.5	P2_r1_0.5mm.png	3.5/4.0	
8	P2	1	1.0	P2_r1_1.0mm.png	7.5/8.0	
9	P2	1	1.5	P2_r1_1.5mm.png	12.0/12.5	
10 Extra Extra	P2 P2 P2	1 1 1	2.0 2.5 3.0	P2_r1_2.0mm.png P2_r1_2.5mm.png P2_r1_3.0mm.png	17.0/17.5 22/22.5 28.0	
11	P3	1	ref	P3_r1_ref.png	0.5	
12	P3	1	0.5	P3_r1_0.5mm.png	1.5/2.0	
13	P3	1	1.0	P3_r1_1.0mm.png	3.0/3.5	
14	P3	1	1.5	P3_r1_1.5mm.png	5.0/5.5	
15 Extra	P3 P3	1 1	2.0 2.5	P3_r1_2.0mm.png P3_r1_2.5mm.png	6.5/7./7.5 9.0/9.5	

#	Piece	Repeat	Depth (mm)	File name	Force (N)	Notes
Extra	P3	1	3.0	P3_r1_3.0mm.png	11.5/12	
Extra	P3	1	3.5	P3_r1_3.5mm.png	14.0/14.5	
Extra	P3	1	4	P3_r1_4.0mm.png	16.5/17	
Extra	P3	1	4.5	P3_r1_4.5mm.png	19.5/20	
Extra	P3	1	5	P3_r1_5.0mm.png	23/23.5	
16	P4	1	ref	P4_r1_ref.png	0.5	
17	P4	1	0.5	P4_r1_0.5mm.png	1.0/1.5	
18	P4	1	1.0	P4_r1_1.0mm.png	2.5/3	
19	P4	1	1.5	P4_r1_1.5mm.png	3.5/4	
20	P4	1	2.0	P4_r1_2.0mm.png	5.0/5.5	
	Extra	1	2.5	P4_r1_2.5mm.png	6.5/7	
	Extra	1	3.0	P4_r1_3.0mm.png	8.0/8.5	
	Extra	1	3.5	P4_r1_3.5mm.png	10.5/11	
	Extra	1	4.0	P4_r1_4.0mm.png	12.5	
	Extra	1	4.5	P4_r1_4.5mm.png	14/14.5	
	Extra	1	5.0	P4_r1_5.0mm.png	17/17.5	
	Extra	1	5.5	P4_r1_5.5mm.png	19.5/20	
	Extra	1	6.0	P4_r1_6.0mm.png	22/22.5	
	Extra	1	6.5	P4_r1_6.5mm.png	25.5/26	
	Extra	1	7	P4_r1_7mm.png	29/29.5	

#	Piece	Repeat	Depth (mm)	File name	Force (N)	Notes
21	R	1	ref	`R_r1_ref.png`		
22	R	1	0.5	`R_r1_0.5mm.png`		
23	R	1	1.0	`R_r1_1.0mm.png`		
24	R	1	1.5	`R_r1_1.5mm.png`		
25	R	1	2.0	`R_r1_2.0mm.png`		
26	P1	2 (repeatability check)	ref	`P1_r2_ref.png`		
27	P1	2 (repeatability check)	0.5	`P1_r2_0.5mm.png`		
28	P1	2 (repeatability check)	1.0	`P1_r2_1.0mm.png`		
29	P1	2 (repeatability check)	1.5	`P1_r2_1.5mm.png`		
30	P1	2 (repeatability check)	2.0	`P1_r2_2.0mm.png`		

Part B — Pieces alone, no skin (force only)

Put the piece on a rigid surface (stand base or metal plate). Touch at ~0.05 N, zero the distance, then go down step by step; stop at ~30 N here too. No photos. Slope force/depth = real stiffness of the piece.

#	Piece	Depth (mm)	Force (N)	Notes
31	P1	0.5		
32	P1	1.0		
33	P1	1.5		
34	P1	2.0		
35	P2	0.5		
36	P2	1.0		
37	P2	1.5		
38	P2	2.0		
39	P3	0.5		
40	P3	1.0		
41	P3	1.5		
42	P3	2.0		

#	Piece	Depth (mm)	Force (N)	Notes
43	P4	0.5		
44	P4	1.0		
45	P4	1.5		
46	P4	2.0		